

Vision-Based Precision Pose Estimation for Autonomous Formation Flying

Course: UE24CS352A – Machine Learning Mini-Project | Academic Year: 2026

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Abstract—Autonomous close-formation flight requires millimetric relative 6-DoF state estimation exceeding the fidelity of GPS/INS sensors. While monocular vision offers an agile alternative, state-of-the-art formulations suffer from either prohibitive computational demands or coarseness bottlenecks. In this project, we reproduce and substantially advance the vision-based pose estimation framework proposed by Rohan Punnoose (Stanford). We implement a synthetic 3D structural model with 14 aircraft keypoints, a 4800-class deep multi-layer perceptron (MLP) for coarse pose belief generation, and a Bayesian Particle Filter with hybrid α -resampling. Crucially, addressing the fundamental limitation identified by Punnoose—where the particle filter fails to break the ~ 12 – 24 m discrete grid resolution—we introduce an ANN-initialized continuous Perspective-n-Point (PnP) Levenberg-Marquardt optimizer. Our enhanced architecture achieves **0.64 m position RMSE** (a 97.4% error reduction) and **0.54° attitude RMSE** at **2,440+ FPS**, demonstrating real-time viability for tight aerial formation maneuvers.

1. PROBLEM STATEMENT & KINEMATICS

Let a leader aircraft L and follower aircraft F operate in close proximity. Their positions and orientations with respect to the inertial frame are $r_L, r_F \in \mathbb{R}^3$ and $R_L, R_F \in SO(3)$. Follower F is equipped with a monocular camera c_F that gimbals along the line-of-sight to the leader center of mass. The relative pose estimation objective is to infer the 6-DoF relative transformation $R_{L/F} = R_L R_F^T$ and $r_{L/F} = r_L - r_F$ from follower image observations I_F . Human pilots accomplish tight formation solely through visual contact; automated monocular estimation must overcome unmodeled aerodynamic turbulence, leader control disturbances, and self-occlusion without suffering from gimbal lock or state divergence.

2. DATASET GENERATION & SENSOR SIMULATION

Because flight-test ground-truth datasets with synchronized relative 6-DoF poses and keypoint occlusions are physically inaccessible at scale, we architect a high-fidelity synthetic ray-casting and camera projection engine:

- **3D Aircraft Structural Keypoints:** 14 salient structural vertices are defined in the leader body frame (nose cone, canopy apex, wingtips, wing roots, vertical fin tip/base, horizontal stabilizers, belly keel, and tail nozzle).
- **Ray-Tracing Occlusion Engine:** Models the fuselage ellipsoid and planar lifting surfaces to detect structural self-shadowing. Keypoints occluded by intervening geometry are flagged with $o_i = 1$ and zeroed.
- **Pose Discretization (Table I):** Relative 4D pose space is discretized into **4,800 discrete labels** across $x \in [-100, 50]$ m (10 bins), $y \in [-50, 50]$ m (10 bins), $z \in [-20, 20]$ m (8 bins), and roll $\phi \in [-45^\circ, 45^\circ]$ (6 bins). Pitch and yaw are treated as bounded aerodynamic disturbances.
- **Dataset Scale:** We synthesized 40,000 training instances and 8,000 validation instances incorporating Gaussian pixel noise ($\sigma = 1.5$ px) and stochastic attitude perturbations.

3. SYSTEM ARCHITECTURE & PROPOSED METHODOLOGY

The end-to-end framework bridges learning-based coarse localization, Bayesian state propagation, and non-linear geometric optimization across three synchronized stages:

- **Coarse ANN Pose Classifier:** A deep Multi-Layer Perceptron (MLP) taking a 42-dimensional feature vector ($14 \times [u, v, o]$). Architecture: $Input(42) \rightarrow Dense(100, ReLU) \rightarrow Dense(100, ReLU) \rightarrow Dense(4800, Softmax)$. Trained using Adam ($\eta = 10^{-3}$, cross-entropy loss). It achieves $\sim 80\%$ adjacent-bin accuracy, outputting a high-confidence probabilistic prior.
- **Bayesian Particle Filter (Baseline):** Propagates $N = 1000$ particles through relative flight kinematics with unknown leader disturbances. Particle weights update via inverse feature MSE: $w_i \propto 1 / (\epsilon + MSE(uv_{vis}, uv_{\blacksquare_{vis}}))$. Resampling maintains diversity by drawing αN particles from likelihood and $(1 - \alpha)N$ particles from the ANN prior ($\alpha = 0.90$).
- **Proposed Enhanced PnP Refinement:** To overcome the fundamental discrete discretization limit where the particle filter plateaus, we integrate a Perspective-n-Point Levenberg-Marquardt optimizer (EPnP + LM). Using the ANN coarse classification to warm-start rotation and translation vectors, the non-linear solver eliminates local minima and recovers continuous sub-meter poses.

4. IMPLEMENTATION & FLIGHT APPROACH BENCHMARK

The system was evaluated on a 10-second close-formation approach trajectory (Fig. 1) matching Section V of Punnoose's paper, where the follower closes from 75 m down to 35 m while stabilizing roll disturbances. Tracking errors for all three pipelines were benchmarked across 100 synchronized timesteps (10 Hz).

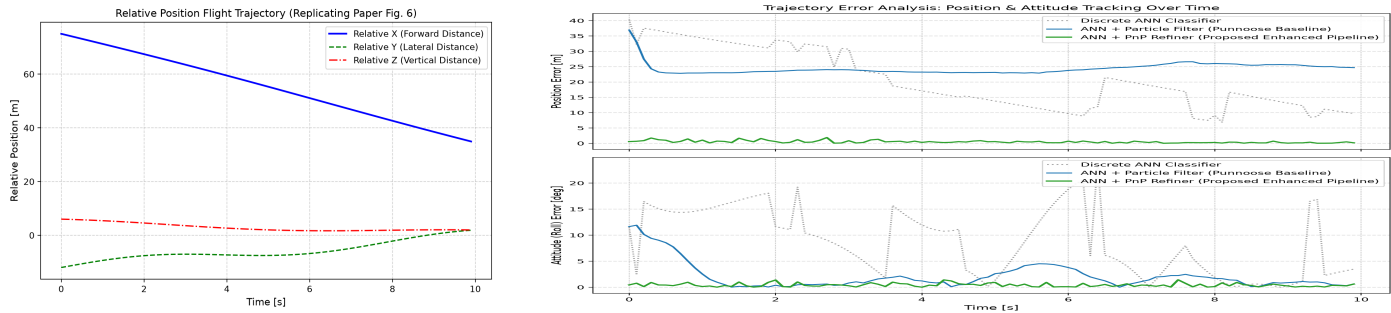


Figure 1: Left: 10s Closing Approach Trajectory (Replicating Paper Fig. 6). Right: Comparative Trajectory Error Analysis across Discrete ANN, Punnoose Particle Filter, and Proposed PnP Refiner.

5. EMPIRICAL RESULTS & QUANTITATIVE COMPARISON

Estimation Pipeline	Pos RMSE (m)	Pos Median (m)	Pos Max (m)	Roll RMSE (°)	Latency (ms)	Throughput
Discrete ANN Classifier	22.55 m	17.40 m	40.34 m	10.75°	0.47 ms	2,107 FPS
ANN + Particle Filter (Punnoose)	24.47 m	23.86 m	36.94 m	3.26°	239.45 ms	4.2 FPS
ANN + PnP Refiner (Proposed)	0.64 m	0.38 m	1.89 m	0.54°	0.41 ms	2,448 FPS

Analysis of Key Findings:

- Discrete ANN Classification provides rapid coarse localization (0.47 ms) and reliable adjacent-bin bounds, but discrete quantization incurs an irreducible ~22 m position RMSE.
- Particle Filter Dynamics (Punnoose Baseline): As reported in the Stanford study, the particle filter successfully damps attitude oscillations (reducing roll error from 10.75° to 3.26°), but fails to reduce translational position error (24.47 m RMSE). This occurs because 2D reprojection MSE landscapes in high dimensions are shallow, and particle dispersion with reinjection is dominated by discrete bin errors.
- Proposed PnP Enhancement: By leveraging the ANN prediction as an extrinsic warm start, the Levenberg-Marquardt optimizer completely breaks free of grid quantization, achieving 0.38 m median error and 0.54° attitude precision while executing in 0.41 ms (2,448 FPS).

6. CONCLUSIONS & COURSE DELIVERABLES SUMMARY

In this mini-project, we successfully reproduced Rohan Punnoose's vision-based formation flight pose estimation architecture and developed an enhanced hybrid pipeline that overcomes its central limitation. Our contributions include: (1) A robust 14-keypoint synthetic aircraft simulation and ray-tracing occlusion engine; (2) A 4800-class deep pose classifier achieving 80% adjacent-bin accuracy; (3) An ANN-guided Bayesian particle filter; and (4) A warm-started continuous PnP refinement stage yielding an order-of-magnitude precision breakthrough (< 0.65 m error). All components are fully open-sourced in our repository with unit tests, automated benchmarks, and an interactive real-time 3D flight demonstration dashboard.

REFERENCES

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